

MSc. in Computing Practicum Approval Form

Section 1: Student Details

Project Title:	Evaluating Hybrid Semantic–Lexical Retrieval Using Voyage-law-2 and SPLADE for EU Climate Legal Documents
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Chosen major:	Artificial Intelligence
Supervisor	Prof. Mohammed Amine Togou
Date of Submission	20/11/2025

Section 2: About your Practicum

What is the topic of your proposed practicum? (100 words)

This practicum project tests whether combining semantic embeddings with lexical expansion improves retrieval from dense, formal legal texts, specifically EU climate and environmental law documents. We will build a Retrieval-Augmented Generation (RAG) pipeline that pairs Voyage-law-2 semantic embeddings with SPLADE lexical expansion, and compare it to BM25 and dense-only baselines. The core aim is: can a hybrid retriever surface precise, statute-level passages that are critical for legal reasoning about climate obligations and compliance? We'll measure retrieval quality, factual grounding of generated outputs, and computational cost to determine whether hybrid retrieval is a pragmatic approach for EU climate-law tasks.

Please provide details of the papers you have read on this topic (details of 5 papers expected).

1. Rao, A., Alipour, H., & Pendar, N. (2025). Rethinking Hybrid Retrieval: When Small Embeddings and LLM Re-ranking Beat Bigger Models. arXiv preprint arXiv:2506.00049.

This paper presents a comparison of embedding models in tri-modal hybrid retrieval

for Retrieval-Augmented Generation (RAG) systems. They investigate the fusion of dense semantic, sparse lexical, and graph-based embeddings, focusing on the performance of the MiniLM-v6 and BGE-Large architectures. Contrary to conventional assumptions, their results show that the compact MiniLM-v6 outperforms the larger BGE-Large when integrated with LLM-based re-ranking within the tri-modal hybrid framework. Experiments conducted on the SciFact, FIQA, and NFCorpus datasets demonstrate significant improvements in retrieval quality with the MiniLM-v6 configuration. The performance difference is particularly pronounced in agentic re-ranking scenarios, indicating better alignment between MiniLM-v6's embedding space and LLM reasoning. Their findings suggest that embedding model selection for RAG systems should prioritize compatibility with multi-signal fusion and LLM alignment, rather than relying solely on larger models. This approach may reduce computational requirements while improving retrieval accuracy and efficiency.

2. Mukund, & Easwarakumar. (2025). Optimizing Legal Text Summarization Through Dynamic RAG and Domain-Specific Adaptation. *Symmetry*.

This paper proposes a Dynamic Legal Retrieval-Augmented Generation (RAG) framework to improve legal text summarization, addressing the complexity and specificity of legal language. The system uses a BM25 retriever with top-3 chunk selection to ensure relevant, non-redundant legal context. A compression ratio constraint (0.05–0.5) maintains balance between the original judgment and its summary. Experiments show BM25 provides the best precision–recall tradeoff. Among tested transformer models—DeepSeek-7B, LLaMA 2-7B, and LLaMA 3.1-8B—the LLaMA 3.1-8B model, enhanced with Legal NER and Dynamic RAG, delivers the strongest performance, achieving a BERTScore of 0.89. The study offers guidance for future work in hybrid retrieval methods, adaptive chunking, and legal-specific evaluation metrics, with applications in case law analysis and automated legal drafting.

3. Reuter, Markus & Lingenberg, Tobias & Liepiņa, Rūta & Lagioia, Francesca & Lippi, Marco & Sartor, Giovanni & Passerini, Andrea & Sayin, Burcu. (2025). Towards Reliable Retrieval in RAG Systems for Large Legal Datasets. 17-30. 10.18653/v1/2025.nllp-1.3.

In this paper, they address the hallucination challenge by first identifying and quantifying a critical failure mode called Document-Level Retrieval Mismatch (DRM), where the retriever selects information from entirely incorrect source documents. To mitigate DRM, they investigate a simple and computationally efficient technique which they refer to as Summary-Augmented Chunking (SAC). This method enhances each text chunk with a document-level synthetic summary, thereby injecting crucial global context that would otherwise be lost during a standard chunking process. Their experiments on a diverse set of legal information retrieval tasks show that SAC greatly reduces DRM and, consequently, also improves text-level retrieval precision and recall. Interestingly, they found that a generic summarization strategy outperforms an approach that incorporates legal expert domain knowledge to target specific legal elements. Their work provides evidence that this practical, scalable,

and easily integrable technique enhances the reliability of RAG systems when applied to large-scale legal document datasets.

4. Tang, Yanran & Qiu, Ruihong & Li, Xue & Huang, Zi. (2025). ReaKase-8B: Legal Case Retrieval via Knowledge and Reasoning Representations with LLMs. 10.48550/arXiv.2510.26178.

In this paper, a novel ReaKase-8B framework is proposed to leverage extracted legal facts, legal issues, legal relation triplets and legal reasoning for effective legal case retrieval. ReaKase-8B designs an in-context legal case representation learning paradigm with a fine-tuned large language model. Extensive experiments on two benchmark datasets from COLIEE 2022 and COLIEE 2023 demonstrate that our knowledge and reasoning augmented embeddings substantially improve retrieval performance over baseline models, highlighting the potential of integrating legal reasoning into legal case retrieval systems.

5. Formal, Thibault & Piwowarski, Benjamin & Clinchant, Stéphane. (2021). SPLADE: Sparse Lexical and Expansion Model for First Stage Ranking. 2288-2292. 10.1145/3404835.3463098.

In neural Information Retrieval, ongoing research is directed towards improving the first retriever in ranking pipelines. Learning dense embeddings to conduct retrieval using efficient approximate nearest neighbors methods has proven to work well. Meanwhile, there has been a growing interest in learning sparse representations for documents and queries, that could inherit from the desirable properties of bag-of-words models such as the exact matching of terms and the efficiency of inverted indexes. In this work, they present a new first-stage ranker based on explicit sparsity regularization and a log-saturation effect on term weights, leading to highly sparse representations and competitive results with respect to state-of-the-art dense and sparse methods. Their approach is simple, trained end-to-end in a single stage. They also explore the trade-off between effectiveness and efficiency, by controlling the contribution of the sparsity regularization.

How does your proposal relate to existing work on this topic described in these papers? (200 words)

This project grows out of recent attempts to make information retrieval more reliable for complex, highly structured texts. A recurring finding across current work is that dense models alone, no matter how large, often miss legally important wording. Researchers exploring hybrid systems argue that retrieval becomes stronger when semantic models are paired with methods that still respect exact language. Their results suggest that clever combinations of smaller embedding models and targeted lexical techniques can outperform brute-force scaling. This is exactly the challenge we face with EU climate law: the meaning of an obligation can hinge on a single phrase, so a system that “understands” the law must balance semantic interpretation with strict attention to statutory wording.

Another theme in recent studies is that retrieval failures often stem not from the ranking algorithm itself but from mismatches between what is retrieved and where it came from in the underlying document. Work diagnosing these issues has inspired our inclusion of

document-level context and summary-based chunking to reduce misunderstandings caused by fragmented legal text. Similarly, research showing the value of reasoning-aware retrieval reinforces the idea that legal queries require more than keyword overlap, they require surfacing passages that fit the legal logic of the question. Our proposal builds directly on these insights and tests them in the demanding context of EU climate law.

What are the research questions that you will attempt to answer? (200 words)

The practicum focuses on the following primary research question:

RQ1: How does a hybrid retrieval system combining Voyage-law-2 semantic embeddings and SPLADE lexical expansion perform on EU climate-law documents compared to BM25 and dense-only baselines?

How will you explore these questions? (Please address the following points. Note that three or four sentences on each will suffice.)

- What software and programming environment will you use?

Python, Hugging Face Transformers, LangChain (for orchestration), FAISS or Weaviate for vector search, Elasticsearch (or BM25 implementation) for lexical retrieval, PyPDF / PDFMiner for parsing, and an LLM (e.g., LLaMA-3.1 or GPT-4o-mini) for downstream RAG demonstrations.

- What coding/development will you do?

- Build three retrieval pipelines: dense-only (Voyage-law-2), lexical-only (BM25 / Elasticsearch), and hybrid (Voyage-law-2 + SPLADE fusion).
- Implement chunking and document-level context strategies (e.g., summary-augmented chunking) to reduce Document-Level Retrieval Mismatch.
- Create evaluation scripts for precision@k, recall@k, MRR, retrieval latency, index size, and a factual grounding metric for RAG outputs (automated).

- What data will be used for your investigations?

A curated set of Irish legal documents including Acts, statutory instruments, case summaries, and judicial commentary.

- Is this data currently available, if not, where will it come from?

All materials are publicly available from sources such as the Irish Statute Book, Courts Service publications, and provided internal PDFs. Primary legal datasets and corpora we will use -

a) European Environmental Law, Third Edition.

<https://zagovorniki-okolja.si/wp-content/uploads/2018/11/European-Environment>

[al-Law-Third-Edition.pdf](#)

- b) Sourcebook on EU Environmental Law, European Investment Bank.
https://www.eib.org/files/publications/strategies/sourcebook_on_eu_environmental_law_en.pdf

- What experiments do you expect to run?

- Retrieval effectiveness: compare dense, lexical, and hybrid retrievers on a curated set of climate-law queries.
- Robustness tests: measure Document-Level Retrieval Mismatch rates with standard chunking vs summary-augmented chunking.
- Cost analysis: index size, query latency, and memory trade-offs introduced by SPLADE and hybrid fusion.

- What output do you expect to gather?

Retrieval metrics, RAG accuracy reports, hallucination analyses, and comparative performance summaries.

- How will the results be evaluated?

Results will be assessed using precision@k, recall@k, MRR, contextual accuracy scoring, hallucination rate, latency, and memory usage.